

Utilizing Software to Create and Maintain a RACI Matrix

Project Management through Responsibility Assignment

Why Projects Fail Without a RACI Matrix

Problems:



Multiple people involved

Unclear Role Assignments

Nobody knows who is actually responsible for what. Some tasks have no owner at all. Nobody picks them up because everyone assumes someone else is handling it.

Duplicate Work

Two or more people unknowingly perform the same task, wasting time and resources.

Accountability Confusion

When something goes wrong (which eventually will happen), there's no single person who owns it. Blame starts shifting around the team.

Hidden Decision Paths

The work is done, it's ready to be approved, but nobody knows whose sign-off is actually needed. So, it just waits.

What is a RACI Matrix?

RACI is an acronym. It stands for **R**esponsible, **A**ccountable, **C**onsulted, and **I**nformed. And it maps every single task in your project to one of these four roles for each stakeholder.

R

Responsible

The Doer
Does the actual work for
the task

A

Accountable

The Decision-Maker
Final approval & ownership

C

Consulted

The Advisor
Provides expertise & input

I

Informed

The Recipient
Kept updated on progress

The Project Manager's Perspective

As Project Managers, our job is not just to track tasks - it's to ensure everyone knows exactly where they stand.

01

Define Clear Accountability

Every task needs one owner. Not a team, not a group, just one person who is ultimately responsible for that task being delivered

02

Prioritize Clarity Over Process

Don't add too many roles, too many columns, too much details. The best RACI matrix is the simplest one

03

Establish a Single Source of Truth

It should not be in someone's email, notes, or verbal agreement, it needs to be one document, one place, that everyone refers to

04

Enable Real-Time Visibility

The team should be able to open the document and find the answer in seconds. Anytime, from anywhere

User Needs: What Must the Software Deliver?

From the Project Manager's perspective, the ideal RACI tool must fulfill these core business requirements:

Visual Clarity

The matrix needs to be immediately readable, no manual needed.

Easy Editing

Projects are living things. As they evolve, we need to add or remove roles and tasks in seconds without the layout breaking.

Centralization

Every responsibility assignment lives in one centralized place. Not in emails, not in papers.

Export Capability

At some point you need to communicate this to stakeholders who are not in the tool. You need to be able to generate a clean PDF or XSLX.

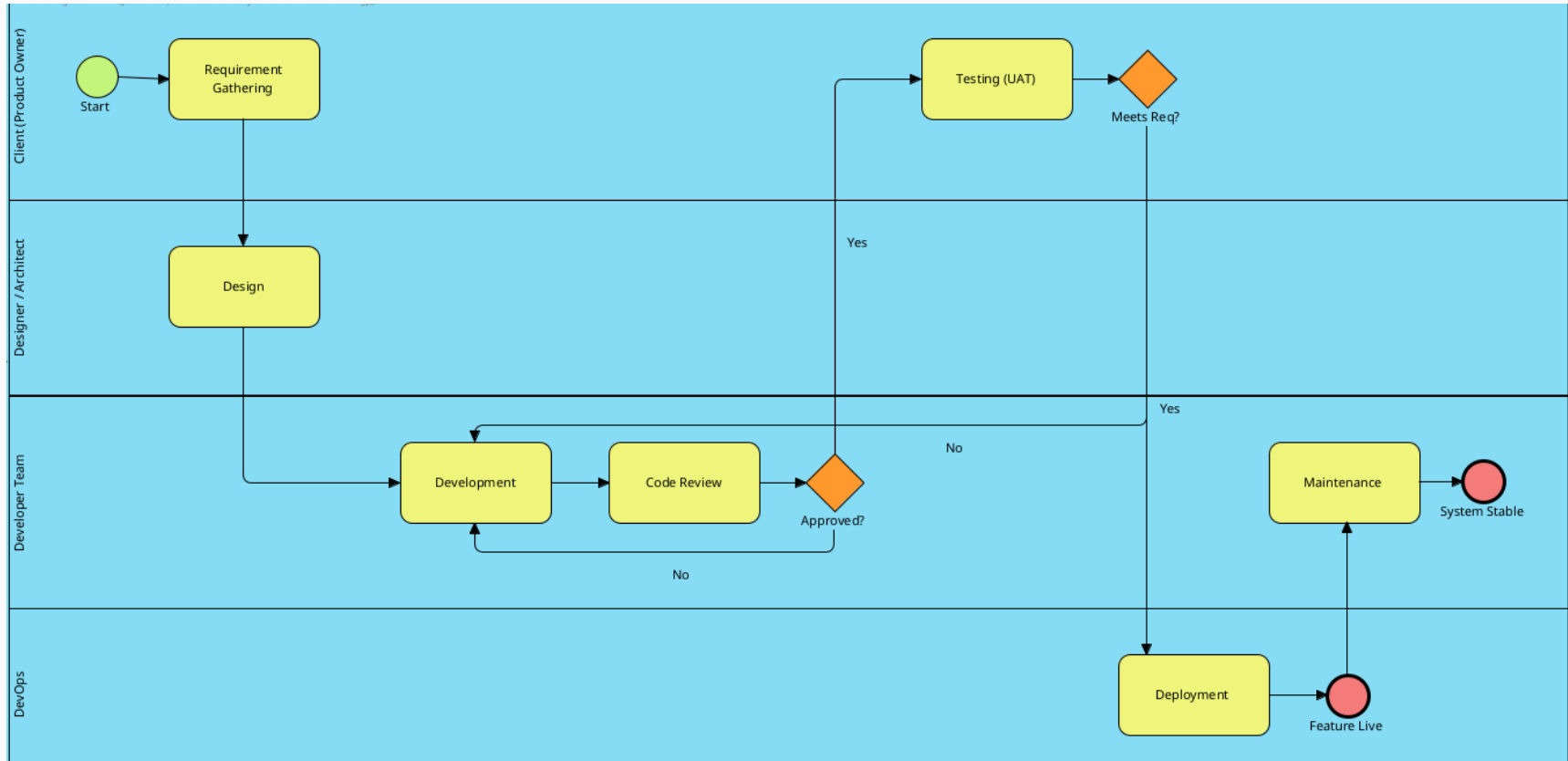
Collaboration

This is a team tool, not a personal one. When someone makes a change, everyone sees it immediately.

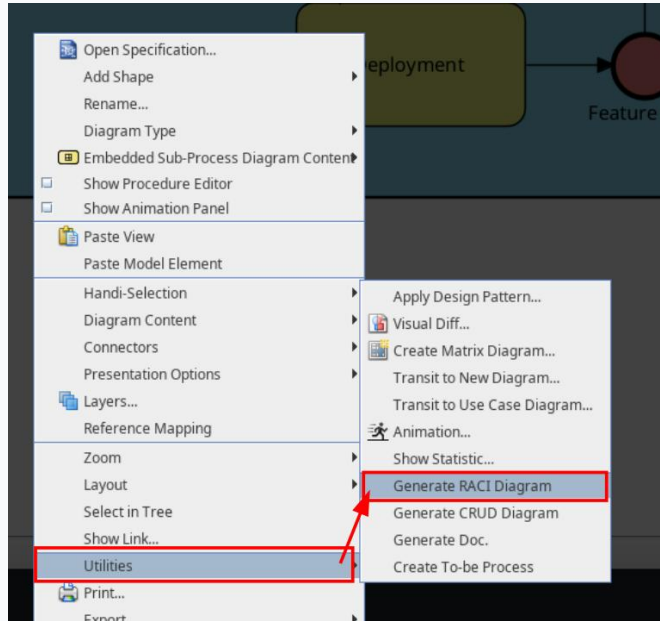
Version Control

This collaboration must be backed by version control where you are able to look back and see what changed, when it changed, and who changed it.

Business Process Model and Notation (Visual Paradigm)



Generate RACI Diagram Automatically (Visual Paradigm)



	Requirement Gathering	Design	Development	Code Review	Testing (UAT)	Deployment	Maintenance
software_development							
Client (Product Owner)	R				R		
Designer / Architect		R					
Developer Team			R	R			R
DevOps						R	

Choosing the right tool

RACI

Project Management through Responsibility Assignment

Functional Specifications (1 of 3): Core Functionality

MoSCoW Method: Must Have | Should Have | Could Have | Won't Have

Functionality	Input	Output	Purpose	MoSCoW
Matrix Grid Editor	Drag/drop roles (columns) & tasks (rows)	Editable 2D RACI matrix	Creates the fundamental RACI structure with R/A/C/I cell assignments	Must Have
Role & Task Management	Add/remove/rename roles and tasks	Updated matrix reflecting all changes	Allows flexibility as project scope and team composition evolve	Must Have
Cell Assignment Logic	Click cell → select R/A/C/I from dropdown	Matrix cell populated with assignment	Enables rapid, consistent responsibility assignment across the team	Must Have
Data Validation Rules	Constraints: each task needs exactly ONE 'A'	Error warnings on invalid assignments	Prevents accountability ambiguity before it becomes a project issue	Should Have

Functional Specifications (2 of 3): Collaboration & Export

MoSCoW Method: Must Have | Should Have | Could Have | Won't Have

Functionality	Input	Output	Purpose	MoSCoW
Real-Time Updates	Multiple users editing the matrix simultaneously	All users immediately see live changes	Team collaborates on assignment decisions without version conflicts	Should Have
Export to PDF/Excel	User selects export format and clicks export	Professional, stakeholder-ready document	Enables formal communication of responsibility assignments	Must Have
Comments on Cells	Hover over assignment, add rationale note	Persistent notes visible on hover/click	Documents WHY each responsibility decision was made	Could Have
Template Library	Select from pre-built RACI templates	Pre-populated matrix for common project types	Accelerates matrix setup for standard project categories	Could Have

Functional Specifications (3 of 3): Integration & Maintenance

MoSCoW Method: Must Have | Should Have | Could Have | Won't Have

Functionality	Input	Output	Purpose	MoSCoW
Version History / Audit Trail	Automatic logging of all edits and changes	Chronological record: who changed what and when	Tracks responsibility evolution and resolves assignment disputes	Should Have
Cloud Storage Sync	Save the matrix; link to Google Drive/OneDrive	Secure, hosted file accessible anytime, anywhere	Prevents data loss and enables remote/distributed team access	Must Have
Integration with PM Tools	Connect to Asana, Monday.com, or Jira task lists	RACI matrix synced with live project tasks	Keeps responsibility assignments aligned with actual execution	Could Have

Key Insight: Cloud storage and basic export are non-negotiable (Must Have). Version history and PM tool integration provide strategic value as the team matures their RACI practices.

Market Analysis: Comparative Tool Evaluation

We evaluated tools based on PM priorities: **ease of use**, real-time collaboration, **export options**, **free tier** availability, and learning curve for a 45-minute lab session.

Tool	Type	Producer	Price (USD)	Best For
Microsoft Excel / Google Sheets	Spreadsheet	Microsoft / Google	Free (For real)	Simple, familiar matrices - no learning curve; limited real-time collaboration
Asana	Project Management	Asana Inc.	Free (2 users) / \$10.99-\$24.99/user/mo	Full PM suite with RACI views - powerful but complex for RACI-only use
Monday.com	Project Management	monday.com Inc.	Free (2 users – 3 boards) / \$9-\$19/user/mo	Visual RACI + workflow automation - feature-rich but steep learning curve
Lucidchart	Diagramming	Lucid Software	Free (3 documents) / \$7.95-\$9/mo	Professional diagram visuals - not designed for responsibility matrices
Miro	Collaboration Canvas	Miro Inc.	Free (3 boards) / \$8-\$16/user/mo	Real-time collaboration canvas - flexible but unstructured for RACI work

Recommended Tool: Google Sheets

★ Selected for: Simplicity + Accessibility + RACI-ready structure + Zero cost to students

WHY GOOGLE SHEETS?

Native grid layout maps directly to RACI structure (roles = columns, tasks = rows)

Real-time collaboration: entire class edits and views simultaneously

Export to PDF and Excel in two clicks - stakeholder-ready output

100% free with any Google account - no credit card, no trial expiry

Students already familiar with spreadsheets - minimal learning curve

Conditional formatting enables color-coded R/A/C/I cells

WHY NOT THE OTHERS?

Excel (Desktop)

No real-time collaboration in free version; requires installation

Asana / Monday.com

Overkill for RACI-only needs; 45 minutes isn't enough to learn the platform

Lucidchart

Designed for diagrams - tables require workarounds; not intuitive for matrix input

Miro

Unstructured canvas makes RACI grids hard to align and read in presentation context

Jira

Developer-focused; requires admin setup and organizational accounts

Getting Started with Google Sheets

A quick overview of Google Sheets as your RACI tool for this lab.

- 1 What is Google Sheets?**

Google Sheets is a free, browser-based spreadsheet application. Its grid layout maps perfectly to a RACI matrix - no installation needed, accessible from any device.
- 2 Why Use It for RACI?**

Real-time collaboration, easy sharing, built-in export to PDF and Excel, and conditional formatting for color-coded R/A/C/I cells - all for free with any Google account.
- 3 How We'll Use It**

In this lab we use the Vertex42 RACI Matrix template modified to our needs (**Shared by us**) - a ready-made spreadsheet you download, open in Google Sheets, fill in, and export. No setup from scratch required.
- 4 Accessing Google Sheets**

Visit sheets.google.com and sign in with any Google or Gmail account. It is completely free - no credit card or trial period required.
- 5 Sharing & Collaboration**

Use the Share button (top right) to invite teammates. Everyone sees changes in real time - ideal for filling in a RACI matrix as a group during class.
- 6 Exporting Your Work**

When finished, use File > Download to export as PDF or Excel (.xlsx). These formats are suitable for submitting your work or sharing with stakeholders.
- 7 Conditional Formatting**

Apply color rules via Format > Conditional formatting to auto-color R/A/C/I cells. This makes your matrix instantly readable at a glance - a key visual best practice.

Exercise - Task 1: Download & Fill the Template

Scenario: Website Redesign Project | Use the Vertex42 RACI Matrix template

YOUR ASSIGNMENT

1. Go to [our shared RACI Matrix.xlsx](#) and download the free RACI Matrix template. **Open it in Google Sheets via File > Import > Upload > Examine my PC**
2. Replace the template's placeholder roles (columns) with: **Project Manager, Designer, Developer, Client**.
Replace placeholder tasks (rows) with: **Define Requirements | Create Mockups | Code Frontend | Code Backend | User Testing | Deploy**
3. Fill in each cell with **R, A, C, or I**. Follow the template's built-in color coding. Rules: **each task needs exactly ONE A, at least one R**, and C/I used only where genuinely needed.
4. Rename your file to Lab6_RACI (click the filename at top). Then add a short comment next to at least two cells explaining your reasoning - use Insert > Comment in Google Sheets.

EXAMPLE: 'Create Mockups' Row

Role	Assignment	Rationale
Designer	R	Does the design work
Project Manager	A	Approves final mockups
Developer	C	Consulted on feasibility
Client	C	Consulted on vision/brand



Pro Tips

- The template already includes color coding - keep it; it makes your matrix instantly readable
- Only ONE 'A' per row - if you want two, decide who truly owns the outcome
- Use comments (Insert > Comment) to justify at least 2 of your assignment choices

Exercise - Task 2: Export & Documentation

Export your completed modified template from Task 1 in multiple formats

1

Export as PDF

In Google Sheets: File → Download →
PDF Document (.pdf)
Make sure "Current sheet" is selected.
Click Export.

2

Export as Excel (.xlsx)

In Google Sheets: File → Download →
Microsoft Excel (.xlsx)
The file downloads instantly - keep it as a
backup of your original template work.

3

Share via Link

Click Share (top right) → Change access
to "Anyone with the link can view" →
Copy the link. Submit this link along with
your exported PDF and Excel files.

Reflection Questions - Add Your Answers as a Comment or Text Box in the Spreadsheet:

Q1: What differences do you notice between the PDF and Excel exports? Which format would you send to a client and why?

Q2: When sharing your matrix with someone who was not in the room, what context would they need that the template alone does not provide? How could you add that context?

Exercise - Task 3: Real-World Application

Individual or Group Activity | Be creative - use a scenario you know!

STEP 1 - Choose Your Scenario

Pick a real situation you are familiar with. You will build a RACI matrix around it from scratch.

Example ideas: Planning a student event | Building a group class project | Organizing a hackathon | Launching a small online store | Running a sports tournament | Setting up a campus club

STEP 2 - Build Your RACI Matrix (Minimum Requirements)

- **Minimum 4 roles** (e.g., Organizer, Sponsor, Volunteers, Participants)
- **Minimum 6 tasks** (plan your scenario comprehensively)
- **Proper R/A/C/I distribution:** Each task has exactly ONE A, most have an R, and C/I are used strategically (not as defaults)

STEP 3 - Document Your Rationale

In a text box or comment column, answer: Why is [Person] accountable for [Task]? Who else was considered? Who needs consultation vs. just notification? (Just for one task of your choice)